

Mathematical Studies Analysis

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1 The fundamental theorem of calculus

1.1 Differentiation of monotone functions

Theorem (Lebesgue's differentiation theorem). Let $I \subset \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be a monotone function. Then f is differentiable a.e. on I .

To show this, we need a few lemmas.

Lemma. Let $I \subset \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let

$$E = \{x \in I^\circ : \text{there exist } f'_+(x) \text{ and } f'_-(x) \text{ and } f'_+(x) \neq f'_-(x)\}.$$

Then E is countable.

Proof. First observe that for any function $g : I \rightarrow \mathbb{R}$, the set of its strict local minimum points in I° is countable. Set

$$E_- = \{x \in E : f'_-(x) < f'_+(x)\}.$$

For every $r \in \mathbb{Q}$, define

$$E_r = \{x \in E_- : f'_-(x) < r < f'_+(x)\},$$

and let $g_r(t) = f(t) - rt$ for $t \in I$. By density of rationals, $E_- = \bigcup_{r \in \mathbb{Q}} E_r$. If $x \in E_r$, then

$$(g_r)'_-(x) < 0 < (g_r)'_+(x).$$

It follows that $g_r(y) > g_r(x)$ for every $y \neq x$ sufficiently close to x , i.e. x is a strict local minimum point of g_r . Therefore E_r is countable and thus E_- is countable. Applying the same argument to $-f$ completes the proof for the lemma. $\square$

Lemma. Let $E \subset \mathbb{R}$ be bounded, $\mathcal{F}$ be family of open intervals such that for all $x \in E$ and $\delta > 0$, there exists $(x, x + h_x) \in \mathcal{F}$ with $0 \leq h_x < \delta$. Then for all $\varepsilon > 0$, there exists disjoint intervals $I_1, \dots, I_m \in \mathcal{F}$ such that

$$\mathcal{L}_o^1\left(E \cap \bigcup_{i=1}^m I_i\right) \geq \mathcal{L}_o^1(E) - \varepsilon, \quad \sum_{i=1}^m \mathcal{L}^1(I_i) \leq \mathcal{L}_o^1(E) + \varepsilon.$$

Proof. *** TO-DO *** $\square$

Definition. For $f : E \rightarrow \mathbb{R}$, define

$$\liminf_{x \rightarrow x_0} f(x) = \sup_{r > 0} \inf_{0 < d(x, x_0) < r} f(x), \quad \limsup_{x \rightarrow x_0} f(x) = \inf_{r > 0} \sup_{0 < d(x, x_0) < r} f(x),$$

Exercise. $\liminf_{x \rightarrow x_0} f(x) = l \in \mathbb{R}$ iff for every $\varepsilon > 0$:

1. there exists $\delta > 0$ such that $l - \varepsilon < f(x)$ for all $x \in E$ with $0 < d(x, x_0) < \delta$.

2. For every $\delta > 0$, there exists $x \in E$ with $0 < d(x, x_0) < \delta$ such that $f(x) < l + \varepsilon$.

Also, $\liminf_{x \rightarrow x_0} f(x) = -\infty$ iff for every $M > 0$ and $\delta > 0$, there exists $x \in E$ with $0 < d(x, x_0) < \delta$ such that $f(x) < -M$.

Now we are ready to show Lebesgue's differentiation theorem. Recall the statement of the theorem below:

Theorem (Lebesgue's differentiation theorem). Let $I \subset \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be a monotone function. Then f is differentiable a.e. on I .

Proof. Define **Dini's derivatives** as follows:

$$\begin{aligned} \overline{D}_+ f(x_0) &= \limsup_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0}, & \underline{D}_+ f(x_0) &= \liminf_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0}, \\ \overline{D}_- f(x_0) &= \limsup_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0}, & \underline{D}_- f(x_0) &= \liminf_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0}. \end{aligned}$$

WLOG assume f is increasing and $I = [a, b]$. Let

$$\begin{aligned} E &= \{x \in I^\circ : \underline{D}_+ f(x) < \overline{D}_+ f(x)\} \\ &= \bigcup_{\substack{r, s \in \mathbb{Q} \\ 0 < r < s}} \{x \in I^\circ : \underline{D}_+ f(x) < r < s < \overline{D}_+ f(x)\} \\ &:= E_{r,s} \end{aligned}$$

It suffices to show that $\mathcal{L}^1(E_{r,s}) = 0$. Let $\varepsilon > 0$ be given.

Let $x \in E_{r,s}$. We have

$$\liminf_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h} < r.$$

Then for each $\delta > 0$, there exists $0 < h_x < \delta$ such that $h_x^{-1}(f(x+h_x) - f(x)) < r$. Let $\mathcal{F}$ be the collection of $(x, x+h_x)$, where h_x is selected as aforementioned while δ ranges over all positive value, and x ranges over $E_{r,s}$. It is then clear that $\mathcal{F}$ satisfies the condition for the previous lemma.

Apply previous lemma, there exists $I_1, \dots, I_m \in \mathcal{F}$ disjoint such that

$$\mathcal{L}_o^1 \left(E_{r,s} \cap \bigcup_{i=1}^m I_i \right) \geq \mathcal{L}_o^1(E_{r,s}) - \varepsilon, \quad (1)$$

$$\sum_{i=1}^m \mathcal{L}^1(I_i) \leq \mathcal{L}_o^1(E_{r,s}) + \varepsilon. \quad (2)$$

Write $I_i = (x_i, x_i + h_i)$, then

$$\sum_{i=1}^n f(x_i + h_i) - f(x_i) < r \sum_{i=1}^n h_i \leq r \mathcal{L}_o^1(E_{r,s}) + r\varepsilon. \quad (3)$$

Now let $V = \bigcup_{i=1}^m (x_i, x_i + h_i)$, which is open. If $x \in E_{r,s} \cap V$, we have

$$s < \limsup_{t \rightarrow 0} \frac{f(x+t) - f(x)}{t}.$$

Then for every $\delta > 0$, there exists f_x such that $0 < t_x < \delta$ and $s < t_x^{-1}(f(x+t_x) - f(x))$. Define $\mathcal{G}$ to be the collection of $(x, x+t_x)$ that is contained in V , where t_x is selected as aforementioned while $\delta > 0$ ranges over all positive value, and x ranges over $E_{r,s} \cap V$.

The previous lemma again gives $J_1, \dots, J_l \in \mathcal{G}$ disjoint such that

$$\mathcal{L}_o^1\left(E_{r,s} \cap V \cap \bigcup_{k=1}^l J_k\right) \geq \mathcal{L}_o^1(E_{r,s} \cap V) - \varepsilon.$$

Write $J_k = (y_k, y_k + t_k)$, we have

$$\begin{aligned} \sum_{k=1}^l f(y_k + t_k) - f(y_k) &\geq s \sum_{k=1}^l t_k \\ &\geq s \mathcal{L}_o^1(E_{r,s} \cap V) - s\varepsilon \\ &\geq s \mathcal{L}_o^1(E_{r,s}) - 2s\varepsilon. \end{aligned} \quad (4)$$

Now compare (3) and (4). Since f is increasing, it is not hard to see that

$$\sum_{i=1}^n f(x_i + h_i) \geq \sum_{k=1}^l f(y_k + t_k) - f(y_k).$$

It follows that

$$\mathcal{L}_o^1(E_{r,s}) \leq \frac{2s+r}{s-r} \varepsilon.$$

Letting $\varepsilon \rightarrow 0$ completes the proof for $\mathcal{L}^1(E_{r,s}) = 0$. This implies that $f'_+(x)$ exists for a.e. $x \in I$. Similarly, $f'_-(x)$ exists for a.e. $x \in I$.

It remains to show the left and right derivatives are finite. ***** TO-DO *****

□